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Accurate thermophysical properties of water using machine-learned potentials

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ABSTRACT

Simulating water from first principles remains a significant computational challenge due to the slow dynamics of the underlying system. Although machine-learned interatomic potentials (MLPs) can accelerate these simulations, they often fail to achieve the required level of accuracy for reliable uncertainty quantification. In this study, we use MACE—an equivariant graph neural network architecture that has been trained using an extensive RPBE-D3 database—to predict density isobars, diffusion constants, radial distribution functions, and melting points. Although equivariant MACE models are computationally more expensive than simpler architectures, such as kernel-based potentials (KbPs), their significantly lower total energy errors allow for reliable thermodynamic reweighting with minimal bias. Our results are consistent with those of previous studies using KbPs; however, equivariant models can be validated against the ground-truth density functional theory (DFT) ensemble with significantly increased efficiency. These findings establish equivariant MLPs as robust and reliable tools for investigating the thermophysical properties of water with DFT-level accuracy.

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I. INTRODUCTION

Water is an important resource for life and a solvent in sustainable energy applications. Justifiably so, computational insights into the structure of water have received large interest and follow a long history of over 50 years.[1–3] Simulations of liquid phase properties of water, particularly below its melting point, are subject to sluggish dynamics requiring significant computational expenses to achieve well-converged statistics.[4–7] A strong effort in the development of empirical force fields tuned to accurately reproduce these properties has led to a selection of performant potentials that accurately reproduce water's liquid state.[8–30]

Similar strides have been taken to repeat these successes from first-principles. The structure of water has been investigated particularly thoroughly using methods based on density functional theory (DFT), often achieving at least qualitative agreement with experiments.[31–47] Computationally, first-principle calculations come at a much higher expense, allowing convergence of statistics only for a subset of observables at low temperatures.[45,48]

Recent work on machine-learned potentials (MLPs) has shown the promise of significantly accelerating materials simulations at little loss of accuracy.[49] Simulations of water using MLPs not only leverage the increased efficiency for a better understanding of water,[50–58] but also act as a benchmark for different MLP architectures or training techniques.[59–65] Water has therefore been explored with a wide variety of MLPs, for example, GAP,[66] HDNNPS,[46,54,67] deep potentials,[55–57,68] Allegro,[69] and MACE.[62,70] The training databases for these potentials were generated using a similarly large spread of DFT functionals, including PBE,[71] SCAN,[55,57,68] revPBE,[46,70] and RPBE.[52,54,63–65,72] This has resulted in a range of predictions of important quantities, such as densities, diffusion coefficients, and melting points, sometimes varying even when starting from the same or similar DFT ground truth.[63,72] Differentiating between these predictions is not trivial, as the uncertainty of

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a machine learning approach is difficult to evaluate. Computing energy or force errors of some subset of structures does not generally suffice to guarantee accurate observables.[61]

Thermodynamic reweighting is a technique widely used in Monte Carlo simulations to reweight an observable measured in one ensemble to another at a different temperature or Hamiltonian.[73,74] The approach readily extends to molecular dynamics simulations[75,76] and can be employed to reweight an observable computed by an MLP back to the ground-truth DFT ensemble.[52,77–79] The reweighted prediction is, in principle, equal to the DFT result, despite the uncertainties introduced by the machine learning method. While this approach requires some additional DFT calculations and, therefore, incurs some additional computational cost, it is still significantly cheaper than direct DFT simulations. However, reweighting can only be performed with statistical significance if the sample and target distributions overlap sufficiently. If this is not the case, the reweighted result will disproportionally depend on very few outliers, which in practice cannot be sampled enough to yield reproducible results. In the context of MLPs, this requires the overlap of the MLP and DFT Boltzmann distribution to be large, which can be achieved only if the total energy errors of the MLP are adequately low.

It has been demonstrated across many materials that equivariant graph neural network potentials achieve significantly lower fitting errors than MLPs like GAP, KbPs, and HDNNPs.[80,81] While the increase in accuracy comes with a sizable increase in computational cost, MLPs based on equivariant GNNs still show a significant speed-up compared to DFT. The increased accuracy can be leveraged to accurately reweight measured quantities back to the DFT ground truth with statistical significance.

We employ this approach to differentiate between previous results obtained by training several MLPs on a database generated using the RPBE exchange correlation functional with a Grimme D3 dispersion correction. RPBE-D3 water has been shown to be among the DFT exchange correlation functionals to reproduce experimental results reasonably well and has been extensively studied, with several MLP predictions reported.[52,54,63,64,72] The predictions differ substantially depending on the employed MLP or training database.[52,63,72] While trends reported match qualitatively in all cases, the absolute value of quantities like the density maximum of RPBE-D3 simulated water remains uncertain.

We show that training equivariant MACE potentials leads to MLPs that are accurate enough to enable statistically significant thermodynamic reweighting of the predicted density. The accuracy of these potentials elucidated some previously unnoticed inadequacies in the training database that were remedied. We then calculated the density isobars, diffusion constants, radial distribution functions, and melting points with these MACE potentials. The results obtained here agree well with previously published values obtained using KbPs but could be reweighted using thermodynamic perturbation theory, confirming their validity.

II. METHODS

All DFT calculations were performed using VASP 6.4[82,83] using high accuracy projector augmented wave potentials (O_h and H_h)[84,85] and the RPBE exchange-correlation functional,[86] including a Grimme D3 dispersion correction[87] without Becke–Johnson dampening (zero dampening). For training of the machine-learned potentials, we considered systems containing 64 and 63 molecules using cubic cells. The Brillouin zone was sampled only at the Γ-point. This DFT setup was chosen to be consistent with previous studies.[63,64] Here, all structures were evaluated using a plane wave energy cutoff of 2000 eV. MACE[80,88] models were trained with two different settings: invariant MACE with L = 0 (128 × 0e), and equivariant with L = 2 (128 × 0e + 128 ×1o + 128 × 2e). The number of trainable parameters is 460.000 and 880.000 for the invariant and equivariant model, respectively. Any other hyperparameters were kept equal for both approaches. All models were trained using single-precision only, with loss evaluated on energies, forces, and stresses with weights set to 1, 1000, and 10, respectively. The cutoff radius was set to 6.0 Å to match the cutoff of the KbPs. Models were trained for 250 epochs with a batch size of 1, employing a stochastic weight adjustment to increase the weights of energy and stress terms in the loss function for the last 50 epochs to weights of 1000, 100, and 10 for energies, stresses, and forces. For all other parameters, we refer to our previous publication and the data availability statement.[63]

The training database is based on the water structures generated using active learning of VASP kernel-based potentials (KbP) reported in Ref. 63, recalculated with the setup described earlier. When training equivariant MACE potentials on this database, two inadequacies became evident: first, force errors on new water structures sampled from molecular dynamics (MD) simulations using these MACE potentials showed a systematic increase compared to force errors evaluated on a test set withheld from the original database. This likely implies that MACE yields slightly different structures that are not fully represented using the kernel-based potential and the on-the-fly training relying mostly on the kernel-based potential. Hence, additional structures generated from the MACE ensembles were added to the database. This procedure was repeated until newly drawn test sets showed no meaningful increase in errors. Second, all MACE models showed a non-zero mean shift in their stress predictions with respect to the DFT ground truth. Trial DFT calculations on a subset of 100 structures randomly selected from the training database with an increased plane wave cutoff of 4000 eV confirmed a residual Pulay stress of 50 ± 16 bar. Apart from the increase of the plane wave cutoff up to 4000 eV, the DFT setup was kept equal. This mean stress shift was added *a posteriori* to the diagonals of the stress tensor for all training structures contained in the database. The final database contained 2370 structures, each containing 63 or 64 water molecules, for a total of 452.127 atom entities (and three times as many force components). KbP results reported in this study were retrained with the final database but showed no significant differences from previous results based on the initial data only. This means that the inherent errors of the KbPs somewhat obviate the need for an extremely accurate training dataset, whereas MACE requires very clean data to fully exploit its potential.

Molecular dynamics (MD) simulations used a time step of 1 fs and a hydrogen mass increased to 8 amu to allow for accurate integration of the equations of motion. Diffusion constants were calculated with a hydrogen mass of 1 amu and a time step of 0.5 fs. MD simulations with KbPs were run in VASP 6.4,[82,83] using a Langevin thermostat and a Parinello–Rahman barostat, with all friction constants set 2 ps^{-1}. NpT simulations used a fictitious mass of the lattice of 8000. All MACE simulations in cells containing more than 128 molecules were performed in VASP using entry points to supply

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energies and forces from the MACE ASE calculator[80,89] executed with CUDA equivariance acceleration. The thermostatting settings were identical to the ones used for KbPs. Parallel tempering simulations with MACE models were performed in Lammps[90,91] for four linked replicas using a Nosé–Hoover chains thermostat and an MTK barostat with chain lengths set to 10. The damping constants for the thermo- and barostat were set to 0.01 and 0.1 ps, respectively.

The extreme accuracy of L = 2 equivariant MACE potentials comes with a significant increase in computational cost compared to the KbP simulations. The exact computational cost is somewhat difficult to compare, as the KbPs and MACE are optimized to run efficiently on different hardware. However, even when running single-precision simulations on an RTX6000 using cuda equivariance acceleration, simulations with equivariant MACE models were ~4 times slower than KbP simulations performed using 4 to 16 CPUs and required significantly more memory. With cuda equivariance acceleration, a MACE MD simulation of 128 water molecules with our setup achieved simulation times of 0.8 ns per day. On older GPU architectures, such as NVIDIA P100s, performance drops to 0.25 ns/day. For the system sizes considered here, the VASP KbP requires between 8 and 16 cores to achieve the best performance (see Ref. 63). Even the modest allocation of 8 CPU cores can achieve simulation times of 3.5 ns/day. On massively parallel CPU machines with multiple sockets, this allows one to perform parallel tempering calculations very efficiently, typically allocating 8 cores to each image and performing calculations with, for example, 16 (32) images on a total of 128 (256) cores. This is not possible using MACE, since the available implementation requires one GPU for each image. Consequently, considering the resources available to us, replica exchange calculations were limited to a small number of images for MACE.

III. RESULTS

A. Model evaluation

Compared to KbPs, both invariant and equivariant MACE models yielded significantly lower errors in energies, forces, and stresses for both training and test sets. The equivariant MACE models consistently achieve force root mean square errors of 5 meV/Å with energy errors below 0.1 meV/atom when evaluated on test sets containing structures of the same size as the ones contained in the training database. This is an improvement of up to a factor of 5 compared to the kernel-based approach. To confirm consistent results for systems of varying sizes, we report errors on a separate test set containing structures with 128 water molecules.

These results are somewhat intriguing. First, we observe that the force error for the KbP does not increase from the holdout test set with 63/64 molecules to the 128 atom test set and that the energy per atom error decreases slightly. As discussed previously,[63] root mean square errors (RMSE) *per atom* should decrease as the square root of the number of atoms if there are no long-range correlation effects relevant on the scale of the reported error bars. This means that the RMSE of the *total energy* increases as the square root of the number of atoms, which in turn relates to classical thermodynamics that states that the variance grows linearly with system size in the absence of long-range correlations. However, intriguingly, the energy error for the invariant MACE (L = 0) increases from 64 to 128 atoms, even surpassing the KbP approach. This implies that the invariant MACE has fitted some unphysical long-range effects to the available data. For equivariant MACE (L = 2), we observe a significant improvement; however, there is still a slight increase in error when transitioning from the holdout test set with 63/64 molecules to the evaluation set with 128 molecules. We believe this is due to the long-range interaction between the molecules being mainly dipole-related. Since dipoles are vector-like entities, faithful reproduction requires at least L = 1 equivariant message passing. Our original KbP does not attempt to fit any long-range interactions and, remarkably, this makes it more robust in energy predictions for larger systems, albeit at significantly larger base errors. The KbP approach by construction entirely neglects any electrostatic long-range effects. Equivariant MACE (L = 2) performs significantly better than KbP for both the holdout test set and the larger 128-atom test set, but equivariant MACE still potentially suffers from some inadequacies in extracting long-range electrostatic interactions. However, the increase in errors is so small that we cannot rule out that the error increase is related to the different k-point densities in the 63/64 and 128-molecule ensembles (in both cases, sampling was performed at the Γ-point only).

B. Isobar and density maximum

Density isobars were obtained from parallel tempering simulations with 128 water molecules for at least 10 ns. The results are shown in Fig. 1, with error bars determined by block averaging with a block size of 500. Equivariant MACE models and KbPs show reasonable agreement, particularly at temperatures above the density maximum. The MACE potential predicts a slightly lower density and a slightly greater temperature for the density maximum, but the curves look remarkably similar. Overall agreement between MACE and the KbP and our previous study[63] is very good. The 128 molecule cells have an average box dimension of 14 Å, preventing finite size effects for KbP simulations. The message passing formalism of

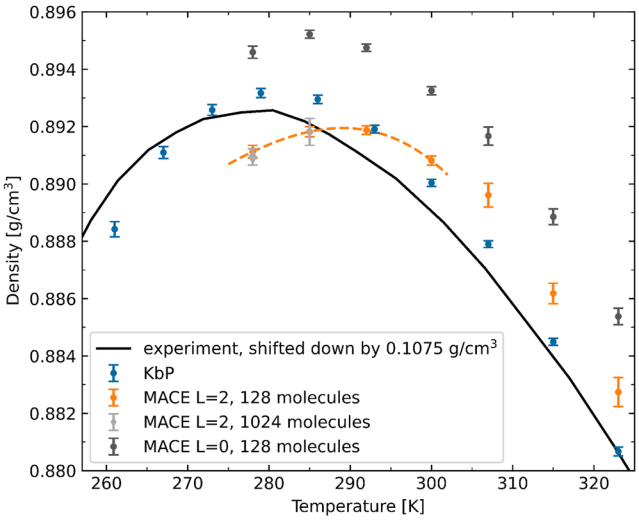


FIG. 1. Density isobars of water for 128 molecule simulations obtained by parallel tempering for KbP and MACE. The data points for 1024 molecules were obtained without parallel tempering but align within uncertainties with the 128 molecule results. The experimental curve was shifted down to facilitate the comparison of the isobars.

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2-layer MACE models results in an effective cutoff distance of 24 Å, larger than any box dimension encountered during the simulations of 128 molecules. To rule out residual finite-size effects due to interactions between repeated images, separate simulations including 1024 molecules were performed at two densities close to the density maximum. At the density maximum, the average length of the lattice vectors in these systems is above 30 Å. Because of the large memory requirement of simulating such large systems, we were not able to perform parallel tempering simulations for 1024 molecules using the resources available to us. However, the results obtained show no indication of a residual finite-size effect.

Despite lower training and test set errors (with the previously noted caveat of larger errors for the total energy for 128 molecules), the results obtained from invariant MACE models appear to be further off from the reference equivariant MACE than the KbP. The shape of the simulated isobar shows a greater deviation from the experimental results than the other potentials. As we established in our tests on 128 molecules, total energy errors are largest for MACE L = 0, so we excluded this potential from our further analysis.

Finally, to control for possible noise introduced by the statistical optimizer, two MACE equivariant models were trained on the final database. The predicted volumes at 325 K agreed to a difference below 0.1%.

The density maximum of RPBE-D3 water fitted with equivariant MACE models was determined by fitting the isobar with a fourth-order polynomial, suggesting that the maximum lies at 288.8 ± 2.4 K. This is in good agreement with the value determined by the KbP, which was 280 and 281 K when calculated based on the old and new databases, respectively.

C. Melting points

The melting point of ice I_h was determined for both potentials by simulating a cell containing a solid–liquid interface at several temperatures. To adequately represent both phases, the simulation cell contained a total of 800 molecules. During the simulations, the internal energies were monitored. A significant increase in the internal energies indicates that the solid part of the cell is melting, and a decrease indicates that the liquid is freezing. We confirmed this by visually inspecting the final structures. The resulting internal energies are displayed in Fig. 2. The melting point was determined to be 283.75 ± 1.0 K. This is 4 K below the melting point determined by KbPs but consistent with the higher temperature of the density maximum determined from the isobar. The temperature difference between the density maximum and the melting point is 6 K, while it was 4 K when simulated with the kernel-based approach. This is in good agreement with the experimental value of 4 K and confirms that RPBE-D3 water reproduces the correct anomalous melting point at temperatures below the density maximum. We even speculate that improved statistics for MACE would bring the temperature difference in closer agreement with the KbP.

D. Diffusion constants and RDF

To calculate the diffusion constants, MD simulations of 1024 molecules were equilibrated in an NpT ensemble for each temperature with a decreased timestep of 0.5 fs and a hydrogen mass of 1. Once the average volume ceased to drift, a structure was extracted from the trajectory with a volume matching the average. The MD

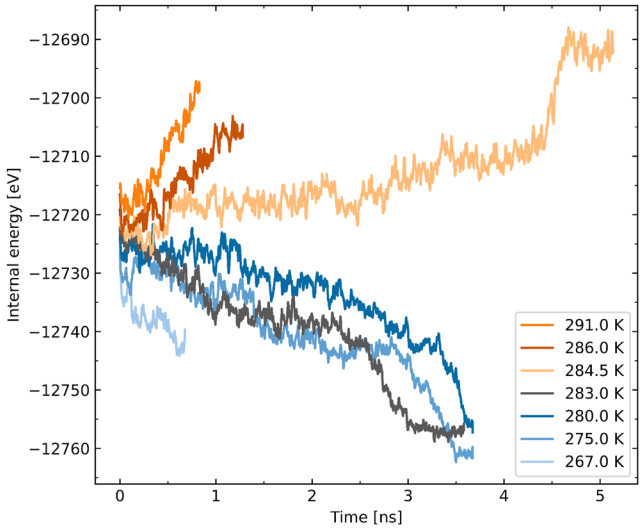


FIG. 2. Internal energies of the direct interface simulations of ice I_h during melting and freezing using the MACE equivariant L = 2 potentials.

simulation was then continued from this starting point in an NVT ensemble. After equilibration, a structure was selected where the total energy equaled the average. The diffusion coefficients were determined from the mean square displacement of an NVE simulation starting from this structure. No significant differences between the MACE potentials and KbPs were observed (Figs. 3–5).

Radial distribution functions were determined from the same trajectories. Again, we show a comparison with the previous KbP

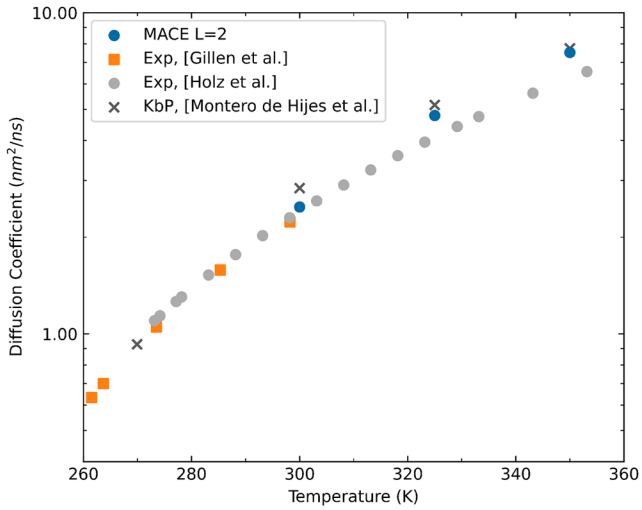


FIG. 3. Diffusion constants obtained from the MACE L = 2 potential. KbP results. Reproduced with permission from Montero De Hijes *et al.*, J. Chem. Phys. 160, 114107 (2024).[63] Copyright AIP Publishing LLC and experimental data from J. Chem. Phys. **57**, 5117–5119 (1972).[92] Copyright AIP Publishing LLC. Experimental data. Reproduced with permission from Holz *et al.*, Phys. Chem. Chem. Phys. **2**, 4740 (2000).[93] Copyright 2000 Royal Society of Chemistry.

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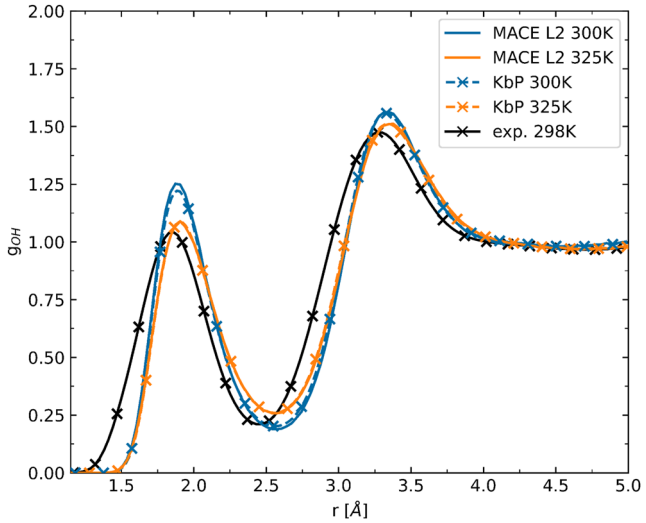


FIG. 4. Hydrogen–oxygen partial radial distribution function at different temperatures, calculated at the respective equilibrium volume with MACE L = 2 potentials. The KbP results. Reproduced with permission from Montero De Hijes *et al.*, J. Chem. Phys. **160**, 114107 (2024).[63] Copyright AIP Publishing LLC.

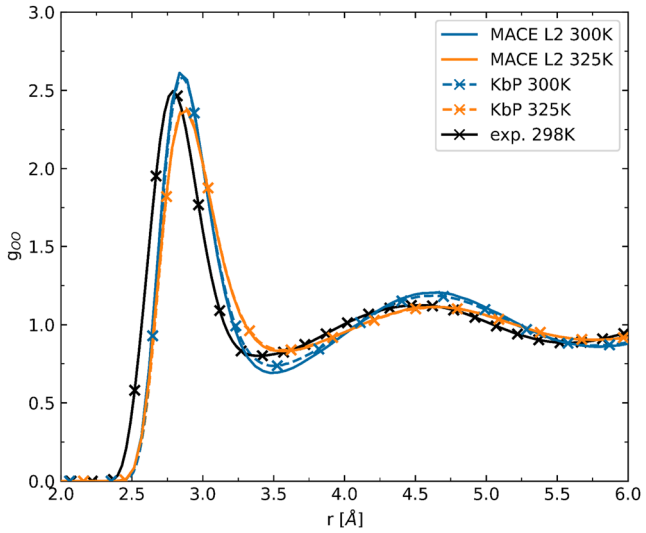


FIG. 5. Oxygen–oxygen partial radial distribution function at different temperatures, calculated at the respective equilibrium volume with MACE L = 2 potentials. The KbP results. Reproduced with permission from Montero De Hijes *et al.*, J. Chem. Phys. **160**, 114107 (2024).[63] Copyright AIP Publishing LLC.

results,[63] showing only minute differences. These are likely related to the different densities at which the simulations were performed. While the diffusion coefficients are in excellent agreement with experimental values, the radial distribution functions show that both potentials are overstructured compared to the experiment. It is possible that part of this error can be attributed to the absence of nuclear quantum effects.[48] Clearly, the classical simulations at T = 325 K are in better agreement with experiment for the heights of the peaks (raising the temperature approximates the nuclear quantum effects to some extent). Furthermore, the RDF peaks are shifted to larger volumes, which relates to the smaller observed densities.

E. Thermodynamic reweighing

In general, the MACE equivariant results agree to within 3% with the results determined by KbPs but show significant differences compared to other potentials like invariant MACE potentials and HDNNPs.[63] All of the results indicate that RPBE-D3 underestimates the density of water by about 10%, with individual predictions of the density maximum varying up to 2% in both density and temperature. To differentiate between the results, we employ thermodynamic reweighting from the MLP ensemble to the ground truth DFT ensemble. For one KbP and one equivariant MACE model, long simulations were performed in a cell containing 63 water molecules at 292 and 323 K. We have chosen a temperature close to the density maximum of the MACE model, as well as a temperature where the density prediction of KbP and MACE potentials differs. The magnitude of the volume fluctuations was similar in all simulations, with a standard deviation in the volume of 2%. To yield a low statistical standard error, 5000 largely uncorrelated structures were selected as a test set, ensuring that the average volume of the sample was similar to the total average over the simulation. Using 5000 structures, we estimate that the error of the predicted mean volume should be below $2/\sqrt{5000} = 0.03$ % if the reweighting is statistically reliable. This is more than sufficient for our present purpose.

Using the DFT energies, the average volumes of the selected sample were reweighted by the exponent of their energy difference between MLP and DFT,

$$\langle V\rangle_{\mathrm{DFT}} \approx \frac{\sum_i^N V(x_i) e^{-\beta[E_{\mathrm{DFT}}(x_i)-E_{\mathrm{MLFF}}(x_i)]}}{\sum_i^N e^{-\beta[E_{\mathrm{DFT}}(x_i)-E_{\mathrm{MLFF}}(x_i)]}} = \frac{\sum_i^N V(x_i) w_i}{\sum_i^N w_i},$$

and the variance is given by

$$\mathrm{Var}\,(V)_{\mathrm{DFT}} \approx \frac{\sum_i^N (V(x_i) - \langle V\rangle_{\mathrm{DFT}})^2 w_i}{\sum_i^N w_i}.$$

For the KbP, the large errors in total energies well above 25 meV lead to poorly distributed weights, as shown in Figs. 6 and 7. We show the sample taken at 292 K only, as both temperatures lead to very similar distributions. Specifically, in Fig. 7, we see two issues. The reweighted density distribution clearly deviates from a Gaussian distribution on the left-hand side. We see that many configurations with an originally small contribution (blue bar chart toward the left) are amplified by attaining a large weight w_i, leading to a strongly frayed distribution at the left. Some of the light gray (orange) bars toward the left correspond to one or very few configurations. Furthermore, the volume varies greatly for a specific fixed $E_{\mathrm{DFT}} - E_{\mathrm{MLFF}}$ so that a clear covariance between $E_{\mathrm{DFT}} - E_{\mathrm{MLFF}}$ and the average volume is difficult to establish. This means that reweighting from KbPs to DFT shows extremely poor statistics because of the outsized effect of very few structures in the sample. Reweighting from MACE to DFT (see Fig. 8) does not reveal these effects. As the error metrics suggested (see Table I), the distribution is about a factor of

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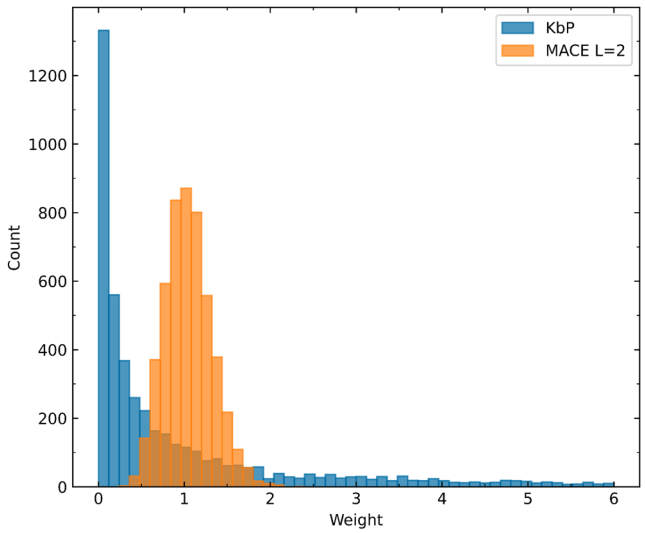


FIG. 6. Distributions of weights w_i obtained for reweighting the KbP and MACE potentials at 292 K. The distribution of weights for the KbP ensemble is heavily skewed, with a large number of samples receiving weights close to zero. The *x* axis is capped at 6.0 for readability. Reweighing the KbP resulted in 520 structures with weights larger than that, up to a maximum of 705.

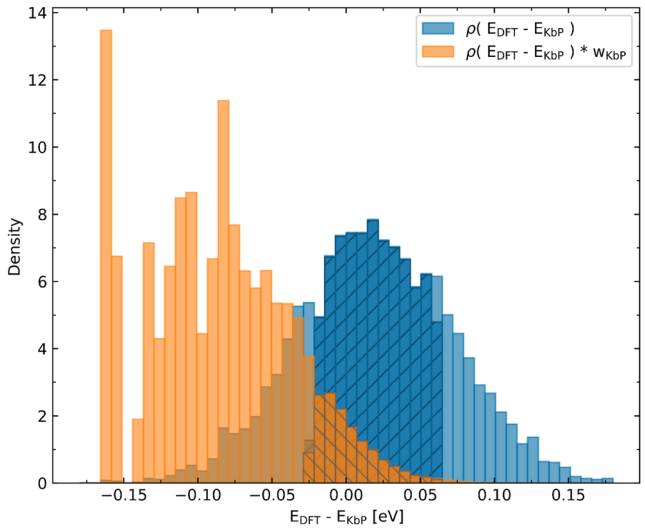


FIG. 7. Distribution of energy differences between KbP and DFT for all sampled structures, along with the distribution multiplied by the weight $\exp[-\beta(E_{DFT} - E_{KbP})]$ at 292 K. The highlighted regions constitute the samples that remain after capping using a capping criterion of 1.72 $k_B T$.

5 narrower for MACE than for KbP. Specifically, the reweighted distribution on the left-hand side of the diagram retains a Gaussian shape, with frays that are no larger than those of the original distribution.

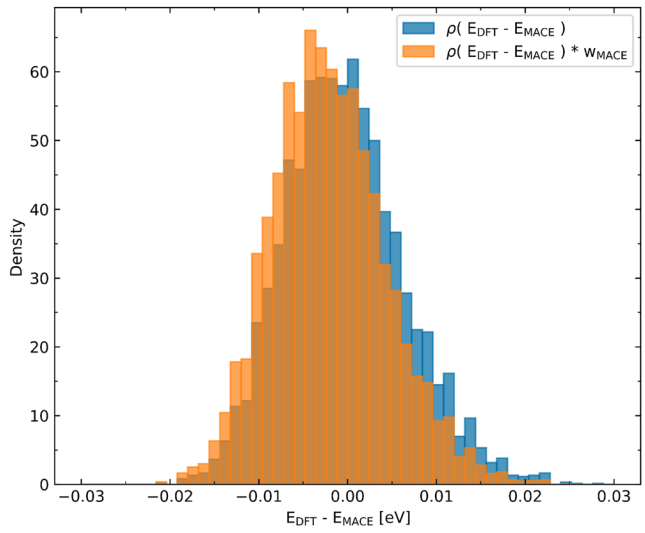


FIG. 8. Distribution of energy differences between MACE and DFT for all sampled structures, along with the distribution multiplied by the weight $\exp[-\beta(E_{DFT} - E_{MACE})]$ at 292 K.

TABLE I. Error metrics for MACE potentials and KbPs. The errors are evaluated on a test set containing 128 molecules drawn from MD simulations of other KbPs. The errors in brackets are holdout test set errors evaluated on structures containing 63 and 64 molecules. The holdout test set structures were removed from the training database before training.

	Force RMSE (eV/Å)	Energy RMSE (meV/atom)
MACE L = 2	8.9(4.9)	0.09(0.07)
MACE L = 0	18.6(10.0)	0.3(0.1)
KbP	27 (26)	0.2(0.3)

The effective sample size N_{eff} quantifies the statistical quality of the average,

$$N_{\text{eff}} = \frac{\left(\sum_i^N w_i\right)^2}{\sum_i^N w_i^2}.$$

The estimated error of the predicted volume is proportional to $\sqrt{\text{Var}\,(V)_{\text{DFT}}/N_{\text{eff}}}$, indicating how a reduction of the effective sample size reduces the statistical significance of the results.

Compared to the parallel tempering simulations performed with 128 molecules, the 63-molecule simulations showed slightly lower densities for both potentials. The results of all thermodynamic reweighting simulations are reported in Table II. For 292 K, the reweighted MACE result of 0.8911 g/cm^3 agrees perfectly with the result obtained from parallel tempering of 0.8914 g/cm^3, while the result obtained from KbPs suggests a lower density of 0.8893 g/cm^3 after reweighting. Reweighting the KbP ensemble shows poor statistics, however. The KbP sample achieved an effective sample size of only 720, raising the effective standard error of the reweighting significantly. The computed standard error of 0.0024 g/cm^3 is large enough that the uncertainty of reweighting spans the density

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TABLE II. Results for the density ρ in g/cm^3 and volume V in Å^3. The results for the ensemble prepared using the machine-learned potential (MACE and KbP) and reweighted ensembles are shown at 292 K. Root-mean-square errors (RMSEs) for the difference between the machine-learned energy and DFT energy are also shown. Estimated errors are calculated from the variance and the effective sample size.

	$\langle\rho\rangle$ (V)	$\langle\rho\rangle_{\text{reweighted}}$ (V)	$\langle\rho\rangle_{\text{reweighted}}$ (V)[a]	RMSE (meV)
MACE L = 2292 K	0.8906 (33.59)	0.8911 ± 0.0004 (33.57)	no change upon capping	6.75
KbP 292 K	0.8903 (33.60)	0.8893 ± 0.0024 (33.64)	0.8898 ± 0.0007 (33.62)	56.50
MACE L = 2323 K	0.8807 (33.97)	0.8813 ± 0.0004 (33.95)	no change upon capping	7.83
KbP 323 K	0.8795 (34.02)	0.8845 ± 0.0033 (33.82)	0.8809 ± 0.0007 (33.96)	65.78

[a]Using capping (i.e., removing) outliers with an energy difference between KbP and DFT energies deviating more than 1.72 and 1.97 k_BT from the mean energy difference for 292 and 323 K, respectively.

results computed by any MLP at this temperature. While it could be inferred that the resulting density is qualitatively correct, the large error prevents quantitative analysis. The equivariant MACE potential yielded well-distributed weights and an effective sample size of 4200, confirming the exceedingly high accuracy of the reweighting in this case. The distributions of the resulting weights are displayed in Fig. 6.

At 323 K, reweighting again leads to a slightly lower density compared to the direct MLP-based simulations. Reweighting the MACE ensemble results only in a minuscule change, with the final result of 0.8813 g/cm^3 agreeing exceptionally well with the direct MLP results. While reweighting the KbP density increases the initially too low result, it overshoots the MACE result to an even larger density of 0.8845 g/cm^3. The increased energy and density error of the 323 K ensemble leads to even worse statistics with an effective sample size of only 70, leading to a standard error of 0.0033 g/cm^3. Again, this error is large enough that other data points, such as the MACE L = 2 result, are included within the estimated range.

"Capping" the weight distributions by excluding structures with out-sized influence can somewhat remedy poor reweighting statistics and increase the effective sample sizes. Specifically, we disregard all structures where the mean removed absolute energy difference $|E_{\text{DFT}}(x_i) - E_{\text{MLFF}}(x_i) - \frac{1}{N}\sum_j^N [E_{\text{DFT}}(x_j) - E_{\text{MLFF}}(x_j)]|$ exceeds a certain capping threshold measured in units of k_BT. In Fig. 7, the shaded region of the histogram highlights the samples remaining after capping the KbP sample at 292 K to achieve the maximum possible effective sample size. While maximizing the effective sample size N_{eff} is a reasonable way to pick the capping criterion, it is by no means the only one. The choice of the capping criterion introduces additional uncertainty and bias if the reweighted average strongly depends on the capping criterion, as is the case for the KbP ensemble. This is shown in Fig. 9. For the KbP sample at 292 K, capping at 1.72 k_BT around the mean energy difference removed 2090 samples and resulted in an effective sample size of 1620, yielding a density of 0.8905 g/cm^3, closer to the MACE results. At 323 K, the highest effective sample size of 1670 was achieved by capping around 1.97 k_BT away from the mean, "removing" 1682 samples and resulting in a density of 0.8809 g/cm^3. For both temperatures, reweighting the equivariant MACE volumes did not alter the obtained results by more than 0.1%, and capping the distribution at any point would decrease its effective sample size. Clearly, reweighing the MACE ensemble is possible with very high statistical significance, and equally important, the results did not change within the statistical error bars from the direct simulation results of MACE.

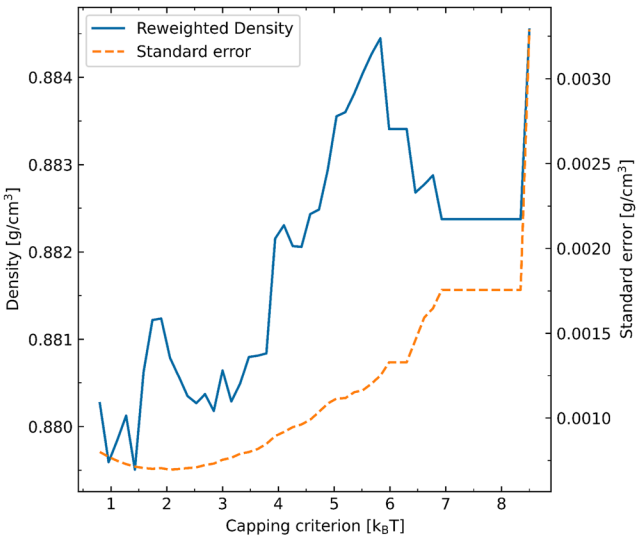


FIG. 9. Reweighted density and predicted standard error of the mean volume obtained for the KbP sample at 323 K vs the chosen capping criterion.

In summary, this analysis suggests that the density isobars simulated with equivariant MACE potential correspond to the DFT isobar to within at least 0.1%, confirming that RPBE-D3 underestimates the density of water by 10.9%. In the KbP potentials case at both temperatures, the reweighting and the frayed reweighted distribution lead to too noisy results to allow predictions with great statistical significance. However, the KbP results underestimate the density compared to MACE, and show an increase in the density upon reweighing. While quantitative analysis is difficult, the trend upon reweighting is correct, with the results obtained after capping matching the ground truth.

IV. CONCLUSION

In this study, we used machine-learned interatomic potentials to determine highly precise thermophysical properties of RPBE-D3 water. We focused on the density isobars (including the density max-

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imum), diffusion constants, radial distribution functions, and the melting point. Equivariant MACE models reproduce the trends previously reported for kernel-based potentials but yield slightly higher temperatures for the density maximum (289 ± 3 K) and the melting point (283.75 ± 0.75 K) than a kernel-based approach. Diffusion constants remain unchanged and are in excellent agreement with experiments. The pair distribution function also remains in excellent agreement with previous calculations. Discrepancies observed in the radial distribution functions compared to experiments confirm that the simulated liquid is overstructured. This is consistent with the observed lower densities: at larger volumes, the minima of the pair correlation function of water become more pronounced, whereas the position and height of the maxima hardly change. Another likely cause for the overstructuring is the neglect of nuclear quantum effects. A common method to approximate nuclear quantum effects is to increase the simulation temperature by about 10%, which indeed improves agreement with room temperature experiments in our case.

Crucially, the low total-energy errors of the equivariant MACE models enable more reliable thermodynamic reweighting of observables, such as the density, back to the DFT ensemble with statistical significance, providing a practical route to validate MLP predictions at the level of ensemble averages. The reweighting analysis confirms that the MACE-derived density at 292 and 323 K is already within ~0.1% of the DFT result for the sampled conditions, whereas the substantially larger total-energy errors of kernel-based potentials lead to poorly behaved weights and prohibit quantitative validation without introducing additional bias through ad hoc procedures such as weight capping. These results clarify that reweighting is not a general remedy for inaccuracies in a potential; rather, it is only reliable if the MLP and DFT ensembles overlap sufficiently, which in practice requires very small total-energy errors.

One of the most intriguing results of the present study is that invariant MACE produces very small fitting errors for systems containing a similar number of molecules to those in the training set. However, when applied to systems with twice as many atoms, invariant MACE performs poorly, with energy errors that even exceed those of the KbP. This suggests that message-passing networks are susceptible to fitting unphysical interactions, preventing accurate extrapolation of finite-size effects. A somewhat similar phenomenon occurs with equivariant MACE (L = 2), but in this case, the energy errors remain significantly lower than those of kernel-based methods. This implies that training datasets for equivariant message-passing networks require much greater care in their construction than for simpler, short-ranged machine-learned potentials. Specifically, we advise including systems of varying size to enable reliable extraction of long-range effects. We note that we did not attempt this in the present study, since to be entirely consistent, we would have had to perform many more calculations for 128 molecules and increase the k-point density for the 63 molecule calculations. Finally, MACE's supreme accuracy enabled us to identify slight inadequacies in our original training set. These were remedied by correcting the residual Pulay stress error—though small, this was still significant enough to degrade the fitting quality—and by adding additional training structures created using MACE.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Tobias Hilpert: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (equal); Software (equal); Visualization (lead); Writing – original draft (equal); Writing – review & editing (equal). **Georg Kresse**: Conceptualization (equal); Funding acquisition (lead); Methodology (equal); Resources (lead); Software (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABLITY

All training and validation datasets produced in this study, as well as input files to the simulations, can be freely accessed at Zenodo under the link https://doi.org/10.5281/zenodo.19482502.

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